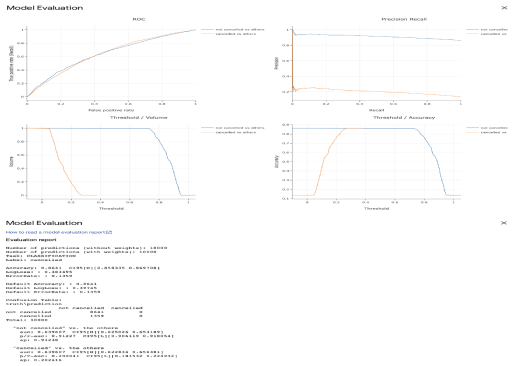
Customer Churn Prediction – Predictive Analytics Project

Overview
Developed and evaluated machine learning classification models to predict customer churn for a regional gym chain. The project used 100,001 training records and 10,001 test records. Models were built using Google Sheets and SimpleML to understand customer behavior and identify drivers of churn.

Key Results
• Model Accuracy: 86.41%
• ROC AUC: 0.6396
• Precision■Recall AUC: 0.9123 (non■churn) / 0.2020 (churn)
• Top predictor: Monthly visit frequency

Business Insights
• Customers with low visit frequency were significantly more likely to cancel memberships.
• Data insights suggested targeted retention programs such as visit incentives and engagement campaigns.
• Demonstrates application of predictive analytics for customer retention strategy.